

Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud.

The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

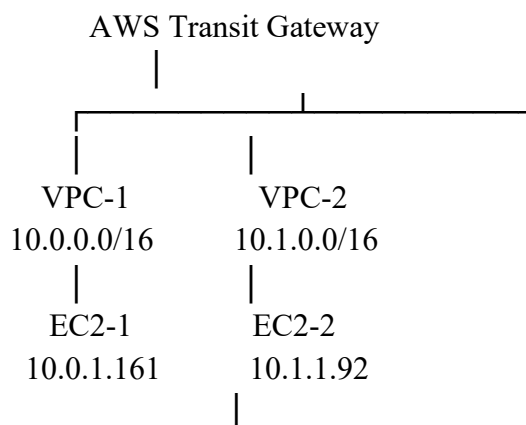
Objectives:

- Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.
- Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

1. Objective

The objective of this task was to configure a multi-VPC AWS network architecture using two VPCs, establish private communication between them using AWS Transit Gateway, and configure an S3 Gateway VPC Endpoint to provide secure access to Amazon S3.

2. Architecture



S3 Gateway Endpoint



Amazon S3

3. Implementation Steps

Step 1: Created VPC-1

Created the first VPC with the CIDR:

10.0.0.0/16

Configured a subnet for EC2-1:

10.0.1.0/24

Launched EC2-1 inside this subnet.

Step 2: Created VPC-2

Created the second VPC with the CIDR:

10.1.0.0/16

Configured a subnet for EC2-2:

10.1.1.0/24

Launched EC2-2 inside this subnet.

The private IP of EC2-2 was:

10.1.1.92

Step 3: Created Transit Gateway

Navigated to:

AWS Console → VPC → Transit Gateways

Created a Transit Gateway to provide centralized connectivity between VPC-1 and VPC-2.

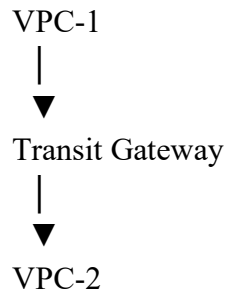
Step 4: Created Transit Gateway Attachments

Created a Transit Gateway VPC attachment for VPC-1.

Created another Transit Gateway VPC attachment for VPC-2.

The two VPCs were therefore connected through the Transit Gateway.

Architecture:



Step 5: Configured VPC-1 Route Table

Added the route for VPC-2:

Destination: 10.1.0.0/16

Target: Transit Gateway

The route table contained:

10.0.0.0/16 → local

10.1.0.0/16 → Transit Gateway

0.0.0.0/0 → Internet Gateway

The Internet Gateway route was configured to allow SSH access to the EC2 instance from my local system.

Step 6: Configured VPC-2 Route Table

Added the route for VPC-1:

Destination: 10.0.0.0/16

Target: Transit Gateway

The VPC-2 route table contained the VPC-to-VPC Transit Gateway route.

This allowed traffic from VPC-2 to reach resources in VPC-1.

4. Transit Gateway Connectivity Test

Connected to EC2-1 using SSH.

EC2-1 private IP:

10.0.1.161

From EC2-1, tested connectivity to EC2-2 using:

ping 10.1.1.92

The ping received successful replies from EC2-2.

Result

64 bytes from 10.1.1.92

The test confirmed that communication between VPC-1 and VPC-2 was successfully established through the Transit Gateway.

Evidence: Screenshot of successful ping and ping statistics.

5. Created Internet Gateway for VPC-2

EC2-2 had a public IP address:

54.169.94.39

Initially, SSH access to EC2-2 was timing out because VPC-2 did not have an Internet Gateway route.

Created and attached an Internet Gateway to VPC-2 and added:

0.0.0.0/0 → Internet Gateway

This allowed SSH access to EC2-2.

6. Created S3 VPC Endpoint

Navigated to:

AWS Console → VPC → Endpoints → Create endpoint

Configured:

Endpoint type: Gateway

Service: Amazon S3

VPC: VPC-2

Created the endpoint:

S3-VPC2-Endpoint

The endpoint became:

State: Available

7. Associated S3 Endpoint with VPC-2 Route Table

Initially, the S3 endpoint did not have a route table associated with it.

Used:

VPC → Endpoints → S3 Endpoint → Manage Route Tables

Selected the VPC-2 route table and associated it with the S3 Gateway Endpoint.

After association, the VPC-2 route table contained an S3 prefix-list route pointing to the VPC endpoint.

This allowed S3 traffic to use the Gateway Endpoint.

8. Configured IAM Permissions

When testing S3 access from EC2-2, the following error was initially received:

Unable to locate credentials

Verified the EC2 IAM role using:

```
aws sts get-caller-identity
```

The EC2 instance was using:

VPCFlowLogsCloudWatchRole

S3 read permissions were added to the role so that the instance could perform the S3 test.

9. Verified IAM Credentials

Executed:

```
aws sts get-caller-identity
```

The command successfully returned the assumed IAM role information.

This confirmed that EC2-2 had valid AWS credentials through its IAM role.

10. Tested S3 Connectivity

From EC2-2, executed:

```
aws s3 ls
```

The command successfully returned the S3 bucket:

devops-vpc-flow-logs-asif-2026

Result

Successful S3 access was confirmed from EC2-2.

This demonstrated that EC2-2 could access Amazon S3 through the configured S3 Gateway VPC Endpoint, subject to IAM permissions.

Evidence: Screenshot showing successful aws s3 ls output.

11. Final Validation

The following validations were successfully completed:

Transit Gateway

ping 10.1.1.92

Result: Successful communication between EC2-1 and EC2-2.

IAM Authentication

aws sts get-caller-identity

Result: IAM role successfully identified.

S3 Access

aws s3 ls

Result:

devops-vpc-flow-logs-asif-2026

Result: Successful S3 access.

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Instances (2) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Filter Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	
☐	EC2-VPC2	i-032e7f865bff4ec93	Running	t3.micro	3/3 checks passed	ap-southeast-1a	-	5
☐	EC2-VPC1	i-02c361bfa7c26adbe	Running	t3.micro	3/3 checks passed	ap-southeast-1a	-	1

All instances

Instance alarm states

In alarm

None

Show instances in alarm

View in CloudWatch

Service health

EC2 service status

Operating normally

View Service Health Dashboard

Health events affecting EC2 instances

Open events

None

Show instances

Upcoming events

None

Show instances

View AWS Health Dashboard

VPC > Transit gateways

Concentrators

Site-to-Site VPN connections

Client VPN endpoints

AWS Verified Access

Verified Access instances

Verified Access trust providers

Verified Access groups

Verified Access endpoints

Transit gateways

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

Traffic Mirroring

Transit gateways (1) Info

Find transit gateway by attribute or tag

	Name	Transit gateway ID	Owner ID	State
☐	TGW-Multi-VPC	tgw-0c63d348002b309bf	331116047691	Available

Select a transit gateway

Search

[Alt+S]

Asia Pacific (Singapore)

Shaik Asif (331116047691)

VPC > Transit gateway attachments

VPC dashboard

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Transit gateway attachments (2)

Find transit gateway attachment by attribute or tag

	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☐	transit gateway attac...	tgw-attach-08fc759928fc46a22	tgw-0c63d348002b309bf	Available	VPC	vpc-0ddeb9e14a40ecbc7
☐	transit gateway attac...	tgw-attach-09975f2d288dcf403	tgw-0c63d348002b309bf	Available	VPC	vpc-0a6826a98209de5b2

Select a transit gateway attachment

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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Privacy

Terms

Cookie preferences

VPC > Route tables > rtb-0e9f4eb8c325fee18

Actions

VPC dashboard

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

rtb-0e9f4eb8c325fee18

Details

Route table ID

rtb-0e9f4eb8c325fee18

VPC

vpc-0a6826a98209de5b2 | vpc -01

Main

Yes

Owner ID

331116047691

Explicit subnet associations

Edge associations

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table
10.1.0.0/16	tgw-0c63d348002b309bf	Active	No	Create Route

Both

Edit routes

ec2-user@ip-10-0-1-161:~

asifs@Asif_Shaik MINGW64 ~/downloads (master)

\$ ssh -i "shaik.pem" ec2-user@18.143.151.64

The authenticity of host '18.143.151.64 (18.143.151.64)' can't be established.

ED25519 key fingerprint is SHA256:W5bDrco+rpha8PbxV1oAFwGvikzP7jGtZbaQIQxBrbA.

This key is not known by any other names

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added '18.143.151.64' (ED25519) to the list of known hosts.



Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

[ec2-user@ip-10-0-1-161 ~]\$ hostname -I

10.0.1.161

[ec2-user@ip-10-0-1-161 ~]\$ ping 10.1.1.92

PING 10.1.1.92 (10.1.1.92) 56(84) bytes of data.

64 bytes from 10.1.1.92: icmp_seq=1 ttl=126 time=3.80 ms
64 bytes from 10.1.1.92: icmp_seq=2 ttl=126 time=0.623 ms
64 bytes from 10.1.1.92: icmp_seq=3 ttl=126 time=0.598 ms
64 bytes from 10.1.1.92: icmp_seq=4 ttl=126 time=0.613 ms
64 bytes from 10.1.1.92: icmp_seq=5 ttl=126 time=0.626 ms
64 bytes from 10.1.1.92: icmp_seq=6 ttl=126 time=0.642 ms
64 bytes from 10.1.1.92: icmp_seq=7 ttl=126 time=0.617 ms
64 bytes from 10.1.1.92: icmp_seq=8 ttl=126 time=0.591 ms
64 bytes from 10.1.1.92: icmp_seq=9 ttl=126 time=0.621 ms
64 bytes from 10.1.1.92: icmp_seq=10 ttl=126 time=0.615 ms
64 bytes from 10.1.1.92: icmp_seq=11 ttl=126 time=0.611 ms
64 bytes from 10.1.1.92: icmp_seq=12 ttl=126 time=0.617 ms
64 bytes from 10.1.1.92: icmp_seq=13 ttl=126 time=0.688 ms
64 bytes from 10.1.1.92: icmp_seq=14 ttl=126 time=0.620 ms
64 bytes from 10.1.1.92: icmp_seq=15 ttl=126 time=0.603 ms
64 bytes from 10.1.1.92: icmp_seq=16 ttl=126 time=0.608 ms
64 bytes from 10.1.1.92: icmp_seq=17 ttl=126 time=0.626 ms
64 bytes from 10.1.1.92: icmp_seq=18 ttl=126 time=0.644 ms
64 bytes from 10.1.1.92: icmp_seq=19 ttl=126 time=0.628 ms
64 bytes from 10.1.1.92: icmp_seq=20 ttl=126 time=0.602 ms
64 bytes from 10.1.1.92: icmp_seq=21 ttl=126 time=0.627 ms
64 bytes from 10.1.1.92: icmp_seq=22 ttl=126 time=0.622 ms
64 bytes from 10.1.1.92: icmp_seq=23 ttl=126 time=0.598 ms
64 bytes from 10.1.1.92: icmp_seq=24 ttl=126 time=0.589 ms
64 bytes from 10.1.1.92: icmp_seq=25 ttl=126 time=0.603 ms
64 bytes from 10.1.1.92: icmp_seq=26 ttl=126 time=0.595 ms
64 bytes from 10.1.1.92: icmp_seq=27 ttl=126 time=0.625 ms
64 bytes from 10.1.1.92: icmp_seq=28 ttl=126 time=0.592 ms
64 bytes from 10.1.1.92: icmp_seq=29 ttl=126 time=0.623 ms
64 bytes from 10.1.1.92: icmp_seq=30 ttl=126 time=0.601 ms
64 bytes from 10.1.1.92: icmp_seq=31 ttl=126 time=0.606 ms
64 bytes from 10.1.1.92: icmp_seq=32 ttl=126 time=7.36 ms

```

64 bytes from 10.1.1.92: icmp_seq=141 ttl=126 time=0.609 ms
64 bytes from 10.1.1.92: icmp_seq=142 ttl=126 time=0.600 ms
64 bytes from 10.1.1.92: icmp_seq=143 ttl=126 time=0.619 ms
64 bytes from 10.1.1.92: icmp_seq=144 ttl=126 time=0.622 ms
64 bytes from 10.1.1.92: icmp_seq=145 ttl=126 time=0.592 ms
64 bytes from 10.1.1.92: icmp_seq=146 ttl=126 time=0.616 ms
64 bytes from 10.1.1.92: icmp_seq=147 ttl=126 time=0.622 ms
64 bytes from 10.1.1.92: icmp_seq=148 ttl=126 time=0.634 ms
64 bytes from 10.1.1.92: icmp_seq=149 ttl=126 time=0.599 ms
64 bytes from 10.1.1.92: icmp_seq=150 ttl=126 time=0.617 ms
64 bytes from 10.1.1.92: icmp_seq=151 ttl=126 time=0.606 ms
64 bytes from 10.1.1.92: icmp_seq=152 ttl=126 time=0.595 ms
64 bytes from 10.1.1.92: icmp_seq=153 ttl=126 time=0.609 ms
64 bytes from 10.1.1.92: icmp_seq=154 ttl=126 time=0.598 ms
64 bytes from 10.1.1.92: icmp_seq=155 ttl=126 time=0.609 ms
64 bytes from 10.1.1.92: icmp_seq=156 ttl=126 time=0.619 ms
64 bytes from 10.1.1.92: icmp_seq=157 ttl=126 time=0.664 ms
64 bytes from 10.1.1.92: icmp_seq=158 ttl=126 time=0.612 ms
64 bytes from 10.1.1.92: icmp_seq=159 ttl=126 time=0.613 ms
64 bytes from 10.1.1.92: icmp_seq=160 ttl=126 time=0.626 ms
64 bytes from 10.1.1.92: icmp_seq=161 ttl=126 time=0.613 ms
64 bytes from 10.1.1.92: icmp_seq=162 ttl=126 time=0.618 ms
64 bytes from 10.1.1.92: icmp_seq=163 ttl=126 time=0.618 ms
64 bytes from 10.1.1.92: icmp_seq=164 ttl=126 time=0.611 ms
64 bytes from 10.1.1.92: icmp_seq=165 ttl=126 time=0.613 ms
64 bytes from 10.1.1.92: icmp_seq=166 ttl=126 time=0.633 ms
64 bytes from 10.1.1.92: icmp_seq=167 ttl=126 time=0.626 ms
64 bytes from 10.1.1.92: icmp_seq=168 ttl=126 time=0.587 ms
64 bytes from 10.1.1.92: icmp_seq=169 ttl=126 time=0.610 ms
64 bytes from 10.1.1.92: icmp_seq=170 ttl=126 time=0.624 ms
64 bytes from 10.1.1.92: icmp_seq=171 ttl=126 time=0.625 ms
64 bytes from 10.1.1.92: icmp_seq=172 ttl=126 time=0.614 ms
64 bytes from 10.1.1.92: icmp_seq=173 ttl=126 time=0.626 ms
64 bytes from 10.1.1.92: icmp_seq=174 ttl=126 time=0.659 ms
64 bytes from 10.1.1.92: icmp_seq=175 ttl=126 time=0.612 ms
64 bytes from 10.1.1.92: icmp_seq=176 ttl=126 time=0.590 ms
64 bytes from 10.1.1.92: icmp_seq=177 ttl=126 time=0.651 ms
64 bytes from 10.1.1.92: icmp_seq=178 ttl=126 time=0.604 ms
64 bytes from 10.1.1.92: icmp_seq=179 ttl=126 time=0.611 ms
64 bytes from 10.1.1.92: icmp_seq=180 ttl=126 time=0.626 ms
64 bytes from 10.1.1.92: icmp_seq=181 ttl=126 time=0.606 ms
64 bytes from 10.1.1.92: icmp_seq=182 ttl=126 time=0.612 ms
64 bytes from 10.1.1.92: icmp_seq=183 ttl=126 time=0.595 ms
64 bytes from 10.1.1.92: icmp_seq=184 ttl=126 time=0.606 ms
64 bytes from 10.1.1.92: icmp_seq=185 ttl=126 time=0.587 ms
64 bytes from 10.1.1.92: icmp_seq=186 ttl=126 time=0.613 ms
64 bytes from 10.1.1.92: icmp_seq=187 ttl=126 time=0.611 ms
^C
--- 10.1.1.92 ping statistics ---
187 packets transmitted, 187 received, 0% packet loss, time 193377ms
rtt min/avg/max/mdev = 0.577/0.668/7.359/0.543 ms
[ec2-user@ip-10-0-1-161 ~]$

```

The screenshot shows the AWS Management Console interface for VPC Endpoints. The left sidebar contains navigation links for Security, PrivateLink and Lattice, DNS firewall, and Network Firewall. The main content area is titled 'Endpoints (1/1)' and shows a table with the following data:

Name	VPC endpoint ID	Endpoint type	Status	Service name
S3-VPC2-Endpoint	vpce-03e1b5485971207a4	Gateway	Available	com.amazonaws.ap-sout

Below the table, the details for the selected endpoint 'vpce-03e1b5485971207a4 / S3-VPC2-Endpoint' are displayed:

- Endpoint ID:** vpce-03e1b5485971207a4
- VPC ID:** vpc-0deba9e14a40ecbc7 (vpc-02)
- DNS record IP type:** service-defined
- Private DNS specified domains:**
- Status:** Available
- Status message:** -
- IP address type:** ipv4
- Creation time:** Tuesday, September 1, 2026 at 15:57:11 GMT+5:30
- Service name:** com.amazonaws.ap-southeast-1.s3
- Service region:** ap-southeast-1
- Endpoint type:** Gateway
- Private DNS names enabled:** No
- Private DNS preference:** -

